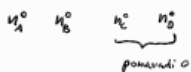
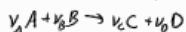


hitrostni zakoni, reakcijski mehanizmi



$$n_A(t) = n_A^0 - \nu_A \xi(t)$$

obseg reakcije, $0 \leq \xi \leq 1$

$$\frac{dn_A}{dt} = -\nu_A \frac{d\xi}{dt} \rightarrow \frac{d\xi}{dt} = -\frac{1}{\nu_A} \frac{dn_A}{dt} = -\frac{1}{\nu_B} \frac{dn_B}{dt} = \frac{1}{\nu_C} \frac{dn_C}{dt} = \frac{1}{\nu_D} \frac{dn_D}{dt}$$

$$\frac{dn_B}{dt} = -\nu_B \frac{d\xi}{dt}$$

$$dn_C = \nu_C \frac{d\xi}{dt}$$

$$dn_D = \nu_D \frac{d\xi}{dt}$$

$$= -\frac{1}{\nu_A} \frac{dC_A}{dt} = -\frac{1}{\nu_B} \frac{dC_B}{dt} = \frac{1}{\nu_C} \frac{dC_C}{dt} = \frac{1}{\nu_D} \frac{dC_D}{dt}$$

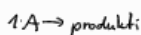
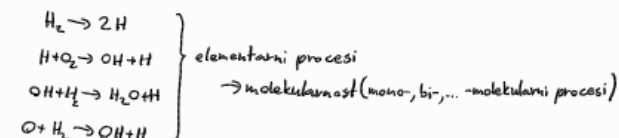
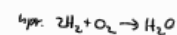
$V = \text{konst.}$

$P(i) \propto C_i$

↓
verjetnost, da se molekula znajde i nahaja na nekem mestu

Če $\Delta_r G < 0$: reakcija je termodinamsko možna (ni pa najpo, da pri danih pogojih steže sama od sebe)

Mehanizmi kem. reakc.



$$-\frac{1}{1} \frac{dC_A}{dt} = k C_A \rightarrow \text{red reakcije}$$

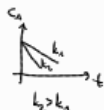
↓
konstanta hitrosti kem. reakc.

$$k, \alpha \in \mathbb{R}$$

1.) Reakcije 0. reda

$$A \rightarrow \text{produkti}$$

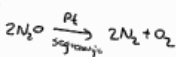
$$-\frac{dC_A}{dt} = k \rightarrow C_A(t) = C_{A0} - kt$$



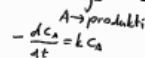
Razpolovni čas

$$t_{1/2} = \frac{1}{k} \frac{C_{A0}}{2}$$

npr.



2.) Reakcije 1. reda



$$-\frac{dC_A}{dt} = k C_A$$

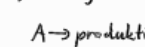
$$\frac{dC_A}{C_A} = -k dt / \int$$

$$\ln \frac{C_A}{C_{A0}} = -kt$$

$$C_A(t) = C_{A0} e^{-kt}$$

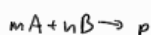
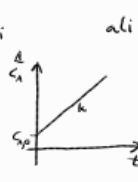
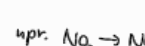


3.) Reakcije 2. reda

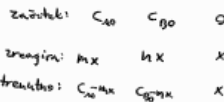


$$-\frac{dC_A}{dt} = k C_A^2$$

$$\frac{1}{C_A(t)} = \frac{1}{C_{A0}} + kt$$



$$-\frac{1}{m} \frac{dC_A}{dt} = -\frac{1}{n} \frac{dC_B}{dt} = k C_A^m C_B^n = \frac{dC_P}{dt}$$



$$\frac{dx}{dt} = k(C_{A0}-mx)(C_{B0}-nx)$$

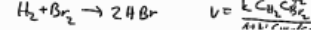
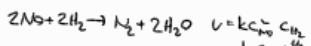
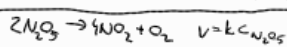
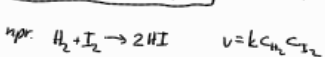
$$\frac{dx}{(C_{A0}-mx)(C_{B0}-nx)} = k dt$$

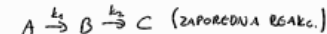
$$\frac{1}{(\dots)(\dots)} = \frac{A}{C_{A0}-mx} + \frac{B}{C_{B0}-nx}$$

$$k t = \frac{1}{m C_{A0} - n C_{B0}} \ln \frac{C_{B0} C_A}{C_{A0} C_B}$$

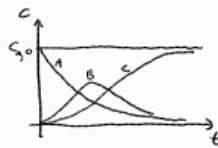
$$A = -\frac{m}{n C_{B0} - m C_{A0}}$$

$$B = \frac{n}{n C_{B0} - m C_{A0}}$$





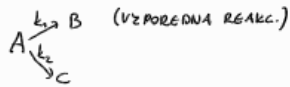
• obe reakciji 1. reda
 $C_A(0) = C_{A0}$; $C_B(0) = C_C(0) = 0$
 $\frac{dC_A}{dt} = -k_1 C_A \rightarrow C_A = C_{A0} e^{-k_1 t}$



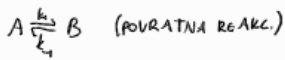
$$\frac{dC_B}{dt} = k_1 C_A - k_2 C_B \rightarrow C_B = \frac{k_1 C_{A0}}{k_2 - k_1} (e^{-k_1 t} - e^{-k_2 t})$$

$$\frac{dC_C}{dt} = k_2 C_B \quad C_C = C_{A0} - C_A - C_B = C_{A0} \left(1 + \frac{k_1 e^{-k_2 t} - k_2 e^{-k_1 t}}{k_2 - k_1} \right)$$

$$C_A + C_B + C_C = C_{A0}$$



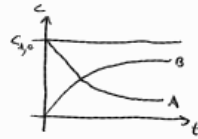
$$\frac{dC_A}{dt} = -(k_1 + k_2) C_A$$



$$\frac{dC_A}{dt} = -k_1 C_A + k_2 C_B \quad C_A = C_{A0} \frac{k_1 + k_2 e^{-(k_1 + k_2)t}}{k_1 + k_2}$$

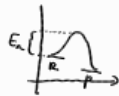
$$\frac{dC_B}{dt} = k_1 C_A - k_2 C_B \quad C_B = C_{A0} \frac{k_1}{k_1 + k_2} \left(1 - \frac{e^{-(k_1 + k_2)t}}{k_1 + k_2} \right)$$

$$C_A + C_B = C_{A0}$$



vpliv T na k
 $\frac{d \ln k}{dT} = \frac{E_a}{RT^2}$ Arrhenius

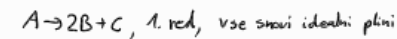
$$\ln \frac{k(T_2)}{k(T_1)} = -\frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$



$$k = A e^{-\frac{E_a}{RT}}$$

↳ deloš faktor, ki imajo dovolj veliko E
 ↳ št. trkov med molekulami

primer:



$$P_{A0} = 0.5 \text{ bar}$$

$$t = 12 \text{ min}$$

$$P_A = 0.1 \text{ bar}$$

$$k, t_{1/2} = ?$$

$$C_A(t) = C_{A0} e^{-kt}$$

	A	B	C
začetek	P_{A0}	0	0
zreagira	$-P_A$	$2P_A$	P_A
koniec	$P_{A0} - P_A$	$2P_A$	$P_A \rightarrow P(t) = P_{A0} - 2P_A$

$$P_A(t_0) = \dots = 0.3 \text{ bar}$$

$$P_A(t_0) = 0.2 \text{ bar}$$

$$t_{1/2} = \frac{\ln 2}{k} = 9.07 \text{ min}$$

$$P_A(t_0) = P_{A0} e^{-kt_0}$$

$$-kt_0 = \log \frac{P_A}{P_{A0}}$$

$$k = \frac{1}{t_0} \log \frac{P_{A0}}{P_A} = 7.65 \cdot 10^{-2} \text{ min}^{-1}$$

primer:



$$E_a = 220 \text{ kJ/mol}$$

$$\begin{cases} T_1 = 20^\circ \text{C} \\ \text{razpade } 30\% A \text{ v } t_1 = 24 \end{cases}$$

$$T_2 = 30^\circ \text{C}$$

$$t_2 = 20 \text{ min}$$

koliko % A ostane?

$$0.7 = e^{-k_1 t_1}$$

$$\rightarrow k_1 = -\frac{1}{t_1} \ln 0.7 = 2.97 \cdot 10^{-3} \text{ min}^{-1}$$

$$\frac{k_2}{k_1} = e^{\frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)}$$

$$\rightarrow k_2 = 5.83 \cdot 10^{-2} \text{ min}^{-1}$$

$$\frac{C_A'}{C_{A0}} = e^{-k_2 t_2} = 0.31$$

$$\rightarrow \text{ostane } \underline{\underline{31\% A}}$$